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Three-dimensional shaping device

Abstract

A three-dimensional shaping device includes: a discharge part that includes a nozzle configured to discharge a shaping material from a nozzle opening thereof; a stage that includes a shaping surface on which the shaping material is to be deposited; a heating part configured to heat the shaping material deposited on the stage; and a control part configured to control the heating part. The heating part includes a first region and a second region closer to an outer periphery of the heating part than is the first region, as viewed from a direction of a perpendicular line of the shaping surface. The control part controls the heating part to make a temperature of the second region higher than a temperature of the first region.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-008462, filed Jan. 24, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a three-dimensional shaping device.

2. Related Art

(3) There is known a three-dimensional shaping device that shapes a three-dimensional shaped

object by discharging and laminating a plasticized material and curing the material.

(4) For example, JP-A-2006-192710 discloses a method for producing a three-dimensional object by extruding a thermoplastic material, which is melted by being heated by a preheater, into a specific region on a base from an extrusion nozzle that performs scanning according to preset shape data, and further laminating the melted material on the material cured on the base.

(5) As described above, when a three-dimensional object is formed by laminating materials, the materials laminated on a base contract due to being cooled down, and thus warpage or deformation occurs in the laminated materials, which may affect the shaping accuracy. In particular, the materials laminated on an outer peripheral side of the base are easily cooled down, and thus are likely to be warped or deformed.

SUMMARY

(6) One aspect of a three-dimensional shaping device according to the present disclosure includes: a discharge part that includes a nozzle configured to discharge a shaping material from a nozzle opening thereof; a stage that includes a shaping surface on which the shaping material is to be deposited; a heating part configured to heat the shaping material deposited on the stage; and a control part configured to control the heating part, the heating part includes a first region and a second region closer to an outer periphery of the heating part than is the first region, as viewed from a direction of a perpendicular line of the shaping surface, and the control part controls the heating part to make a temperature of the second region higher than a temperature of the first region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view schematically illustrating a three-dimensional shaping device according to a first embodiment.

(2) FIG. 2 is a cross-sectional view schematically illustrating the three-dimensional shaping device according to the first embodiment.

(3) FIG. 3 is a perspective view schematically illustrating a flat screw of the three-dimensional shaping device according to the first embodiment.

(4) FIG. 4 is a plan view schematically illustrating a barrel of the three-dimensional shaping device according to the first embodiment.

(5) FIG. 5 is a plan view schematically illustrating a first heating part of the three-dimensional shaping device according to the first embodiment.

(6) FIG. 6 is a perspective view schematically illustrating a second heating part of the three-dimensional shaping device according to the first embodiment.

(7) FIG. 7 is a flowchart illustrating processing performed by a control part of the three-dimensional shaping device according to the first embodiment.

(8) FIG. 8 is a cross-sectional view illustrating shaping-layer formation processing of the three-dimensional shaping device according to the first embodiment.

(9) FIG. 9 is a perspective view schematically illustrating a second heating part of a three-dimensional shaping device according to a second embodiment.

(10) FIG. 10 is a perspective view schematically illustrating a second heating part of a three-dimensional shaping device according to a modification of the second embodiment.

(11) FIG. 11 is a cross-sectional view schematically illustrating a second heating part of a three-dimensional shaping device according to a third embodiment.

(12) FIG. 12 is a cross-sectional view schematically illustrating a second heating part of a three-dimensional shaping device according to a modification of the third embodiment.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(13) Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the drawings. The embodiments described below do not unduly limit contents of the present disclosure described in the claims. Further, all of configurations to be described below are not necessarily essential elements of the present disclosure.

1. First Embodiment

(14) 1.1. Three-Dimensional Shaping Device

(15) 1.1.1. Overall Configuration

(16) First, a three-dimensional shaping device according to a first embodiment will be described with reference to the drawings. FIG. 1 is a perspective view schematically illustrating a three-dimensional shaping device **100** according to the first embodiment. FIG. 2 is a cross-sectional view taken along a line II-II in FIG. 1, schematically illustrating the three-dimensional shaping device **100** according to the first embodiment.

(17) In FIGS. 1 and 2, an X axis, a Y axis, and a Z axis are shown as three axes orthogonal to one another. An X-axis direction and a Y-axis direction are, for example, horizontal directions. A Z-axis direction is, for example, a vertical direction.

(18) As illustrated in FIGS. 1 and 2, the three-dimensional shaping device **100** includes, for example, a discharge part **10**, a stage **20**, a moving mechanism **30**, a support member **40**, a heating part **50**, a first temperature sensor **60**, a second temperature sensor **62**, a third temperature sensor **64**, and a control part **70**. For convenience, illustration of the temperature sensors **60**, **62**, and **64** is omitted in FIG. 1.

(19) The three-dimensional shaping device **100** drives the moving mechanism **30** to change a relative position between the discharge part **10** and the stage **20** while discharging a plasticized shaping material from the discharge part **10** toward the stage **20**. Accordingly, the three-dimensional shaping device **100** shapes a three-dimensional shaped object having a desired shape on the stage **20**.

(20) Although not illustrated, a plurality of discharge parts **10** may be provided. For example, two discharge parts **10** may be provided. In this case, both of the two discharge parts **10** may discharge the shaping material for forming the three-dimensional shaped object, or one may discharge the shaping material and the other may discharge a support material supporting the three-dimensional shaped object.

(21) The discharge part **10** includes, for example, a material supply part **110**, a plasticizing part **120**, and a nozzle **160**.

(22) A material in a form of pellets or powder is fed into the material supply part **110**. The material supply part **110** supplies a material serving as a raw material to the plasticizing part **120**. The material supply part **110** is implemented by, for example, a hopper. The material supplied by the material supply part **110** is, for example, an acrylonitrile butadiene styrene (ABS) resin.

(23) As illustrated in FIG. 2, the material supply part **110** and the plasticizing part **120** are coupled by a supply path **112** provided at a lower side of the material supply part **110**. The material fed into the material supply part **110** is supplied to the plasticizing part **120** via the supply path **112**. In the illustrated example, a “lower side” refers to a $-Z$ -axis direction. An “upper side” refers to a $+Z$ -axis direction.

(24) The plasticizing part **120** includes, for example, a screw case **122**, a drive motor **124**, a flat screw **130**, a barrel **140**, and a barrel heater **150**. The plasticizing part **120** plasticizes the material in a solid state supplied from the material supply part **110**, generates a paste shaping material having fluidity, and supplies the shaping material to the nozzle **160**.

(25) “Plasticizing” is a concept including melting, and refers to changing from a solid state to a state of presenting fluidity. Specifically, in a case of a material in which glass transition occurs, the “plasticizing” refers to setting a temperature of the material to be equal to or higher than a glass transition point. In a case of a material in which the glass transition does not occur, the “plasticizing” refers to setting a temperature of the material to be equal to or higher than a melting

point thereof.

(26) The screw case **122** is a housing in which the flat screw **130** is accommodated. The barrel **140** is provided at a lower surface of the screw case **122**. The flat screw **130** is accommodated in a space defined by the screw case **122** and the barrel **140**.

(27) The drive motor **124** is provided at an upper surface of the screw case **122**. The drive motor **124** is, for example, a servomotor. A shaft **126** of the drive motor **124** is coupled to an upper surface **131** of the flat screw **130**. The drive motor **124** is controlled by the control part **70**.

Although not illustrated, the shaft **126** of the drive motor **124** and the upper surface **131** of the flat screw **130** may be coupled to each other via a speed reducer.

(28) The flat screw **130** has a substantially cylindrical shape in which a size in a direction of a rotation axis R is smaller than a size in a direction orthogonal to the direction of the rotation axis R. In the illustrated example, the rotation axis R is parallel to the Z axis. The flat screw **130** is rotated about the rotation axis R by a torque generated by the drive motor **124**.

(29) The flat screw **130** includes the upper surface **131**, a groove forming surface **132** on an opposite side with respect to the upper surface **131**, and a side surface **133** connecting the upper surface **131** and the groove forming surface **132**. First grooves **134** are formed in the groove forming surface **132**. The side surface **133** is, for example, perpendicular to the groove forming surface **132**. Here, FIG. 3 is a perspective view schematically illustrating the flat screw **130**. For convenience, FIG. 3 illustrates a state in which an up-down positional relationship is reversed from a state illustrated in FIG. 2.

(30) As illustrated in FIG. 3, the first groove **134** is formed in the groove forming surface **132** of the flat screw **130**. The first groove **134** includes, for example, a central portion **135**, a connection portion **136**, and a material introduction portion **137**. The central portion **135** faces a communication hole **146** formed in the barrel **140**. The central portion **135** communicates with the communication hole **146**. The connection portion **136** connects the central portion **135** and the material introduction portion **137**. In the illustrated example, the connection portion **136** is provided extending from the central portion **135** toward an outer periphery of the groove forming surface **132**, presenting a spiral shape. The material introduction portion **137** is provided at the outer periphery of the groove forming surface **132**. That is, the material introduction portion **137** is provided in the side surface **133** of the flat screw **130**. The material supplied from the material supply part **110** is introduced from the material introduction portion **137** into the first groove **134**, passes through the connection portion **136** and the central portion **135**, and is transported to the communication hole **146** formed in the barrel **140**. For example, two first grooves **134** are provided.

(31) The number of the first grooves **134** is not particularly limited. Although not illustrated, three or more first grooves **134** may be provided, or only one first groove **134** may be provided. Although not illustrated, the three-dimensional shaping device **100** may include an in-line screw instead of the flat screw **130**.

(32) As illustrated in FIG. 2, the barrel **140** is provided at a lower side of the flat screw **130**. The barrel **140** has a facing surface **142** facing the groove forming surface **132** of the flat screw **130**. The communication hole **146** communicating with the first groove **134** is formed at a center of the facing surface **142**. FIG. 4 is a plan view schematically illustrating the barrel **140**.

(33) As illustrated in FIG. 4, second grooves **144** and the communication hole **146** are formed in the facing surface **142** of the barrel **140**. A plurality of second grooves **144** are formed. In the illustrated example, six second grooves **144** are formed, and the number of the second grooves **144** is not particularly limited. The plurality of second grooves **144** are formed around the communication hole **146** as viewed from the Z-axis direction. The second grooves **144** each have one end coupled to the communication hole **146** and each extend from the communication hole **146** toward an outer periphery **148** of the barrel **140**, presenting a spiral shape altogether. The second groove **144** has a function of guiding the plasticized shaping material to the communication hole

146.

(34) A shape of the second groove **144** is not particularly limited, and may be, for example, a straight line shape. In addition, one end of the second groove **144** may not be coupled to the communication hole **146**. Further, the second groove **144** may not be formed in the facing surface **142**. However, in consideration of efficiently guiding the plasticized shaping material to the communication hole **146**, the second groove **144** is preferably formed in the facing surface **142**.

(35) As illustrated in FIG. 2, a barrel heater **150** is provided in the barrel **140**. The barrel heater **150** heats the material supplied to between the flat screw **130** and the barrel **140**. An output of the barrel heater **150** is controlled by the control part **70**. The plasticizing part **120** generates a plasticized shaping material by heating the material by the barrel heater **150** while transporting the material toward the communication hole **146** by the flat screw **130**, and the barrel **140**, and causes the generated shaping material to flow out from the communication hole **146**. A shape of the barrel heater **150** may be a ring shape as viewed from the Z-axis direction. The barrel heater **150** may be omitted, and in this case, a heater may be disposed at a position different from the barrel **140**.

(36) The nozzle **160** is provided at a lower side of the barrel **140**. A nozzle flow path **162** is formed in the nozzle **160**. The nozzle flow path **162** communicates with the communication hole **146**. The shaping material is supplied to the nozzle flow path **162** from the communication hole **146**. The nozzle flow path **162** has a nozzle opening **164**. The nozzle **160** discharges the shaping material from the nozzle opening **164** toward the stage **20**.

(37) As illustrated in FIGS. 1 and 2, the stage **20** is provided at a lower side of the nozzle **160**. In the illustrated example, the stage **20** has a shape of a rectangular parallelepiped. The stage **20** has a shaping surface **22** on which the shaping material is to be deposited. The shaping surface **22** is a region of an upper surface of the stage **20**. In the illustrated example, a perpendicular line P of the shaping surface **22** is parallel to the Z axis.

(38) A material of the stage **20** is, for example, a metal such as aluminum. The stage **20** may include a metal plate and an adhesive sheet provided on the metal plate. In this case, the shaping surface **22** is implemented by the adhesive sheet. With the adhesive sheet, the adhesion between the stage **20** and the shaping material discharged from the discharge part **10** can be improved.

(39) Although not illustrated, the stage **20** may include a metal plate in which a groove is formed, and an underlayer provided to fill the groove. In this case, the shaping surface **22** is implemented by the underlayer. A material of the underlayer is, for example, the same as the shaping material. With the underlayer, the adhesion between the stage **20** and the shaping material discharged from the discharge part **10** can be improved.

(40) The moving mechanism **30** supports the stage **20**. In the illustrated example, the moving mechanism **30** supports the stage **20** via a first heating part **170** of the heating part **50**. The moving mechanism **30** changes a relative position between the nozzle **160** and the stage **20**. In the illustrated example, the moving mechanism **30** changes the relative position between the nozzle **160** and the stage **20** in the X-axis direction and the Y-axis direction by moving the stage **20** in the X-axis direction and the Y-axis direction. Further, the moving mechanism **30** changes the relative position between the nozzle **160** and the stage **20** in the Z-axis direction by moving the discharge part **10** in the Z-axis direction.

(41) The moving mechanism **30** includes, for example, a first electric actuator **32**, a second electric actuator **34**, and a third electric actuator **36**. The first electric actuator **32** moves the stage **20** in the X-axis direction. The second electric actuator **34** moves the stage **20** in the Y-axis direction. The third electric actuator **36** moves the discharge part **10** in the Z-axis direction. For example, as long as the relative position between the nozzle **160** and the stage **20** can be changed, the moving mechanism **30** may be configured to move the stage **20** in the Z-axis direction and move the discharge part **10** in the X-axis direction and the Y-axis direction, or may be configured to move the stage **20** or the discharge part **10** in the X-axis direction, the Y-axis direction, and the Z-axis direction.

(42) The support member **40** is coupled to the third electric actuator **36**. The support member **40** supports the discharge part **10** and a second heating part **180** of the heating part **50**. The moving mechanism **30** moves the discharge part **10** and the second heating part **180** in the Z-axis direction by moving the support member **40** in the Z-axis direction by the third electric actuator **36**.

(43) 1.1.2. Heating Part

(44) 1.1.2.1. First Heating Part

(45) The heating part **50** heats the shaping material deposited on the shaping surface **22** of the stage **20**. As illustrated in FIGS. **1** and **2**, the heating part **50** includes the first heating part **170**. The first heating part **170** is provided at a lower side of the nozzle opening **164**. The first heating part **170** is provided at a lower side of the shaping surface **22**. The first heating part **170** is supported by the moving mechanism **30**. The first heating part **170** is provided between the moving mechanism **30** and the stage **20**. The first heating part **170** moves in conjunction with the stage **20**. The first heating part **170** heats the stage **20**.

(46) As illustrated in FIG. **2**, the first heating part **170** includes, for example, a heat insulating member **172**, a lower plate **174**, a heater **176**, and an upper plate **178**.

(47) The heat insulating member **172** is provided on the moving mechanism **30**. The heat insulating member **172** is provided between the moving mechanism **30** and the lower plate **174**. A shape of the heat insulating member **172** is, for example, a plate shape. As the heat insulating member **172**, for example, a ROSLIM™ (registered trademark) board is used. The heat insulating member **172** can reduce the heat of the heater **176** transferred to the lower side of the heat insulating member **172**.

(48) The lower plate **174** is provided on the heat insulating member **172**. The lower plate **174** is provided between the heat insulating member **172** and the heater **176**. A material of the lower plate **174** is, for example, aluminum. An upper surface and a lower surface of the lower plate **174** are, for example, polished mirror surfaces. Accordingly, the lower plate **174** can reflect radiant heat from the heater **176** toward the stage **20**.

(49) The heater **176** is provided on the lower plate **174**. The heater **176** is provided between the lower plate **174** and the upper plate **178**. The heater **176** is fixed by being sandwiched between the lower plate **174** and the upper plate **178**. The heater **176** is not bonded to the lower plate **174** with a bonding agent. The heater **176** is not bonded to the upper plate **178** with a bonding agent. The heater **176** is fixed by its own weight or a load of the upper plate **178**. The heater **176** is, for example, a plate-shaped heater plate. As the heater **176**, for example, a rubber heater is used. The heater **176** heats the stage **20** via the upper plate **178**.

(50) The upper plate **178** is provided on the heater **176**. The upper plate **178** is provided between the heater **176** and the stage **20**. The stage **20** is provided on the upper plate **178**. A material of the upper plate **178** is, for example, aluminum. For example, an oxide film is provided on an upper surface and a lower surface of the upper plate **178**. With the oxide film, radiant heat from the heater **176** can be easily accumulated, and the stage **20** can be efficiently heated. The stage **20** is configured to be detachable. The upper plate **178** can prevent the heater **176** from being exposed when the stage **20** is removed.

(51) Here, FIG. **5** is a plan view schematically illustrating the first heating part **170**. For convenience, illustration of members other than the heater **176** of the first heating part **170** is omitted in FIG. **5**.

(52) As illustrated in FIG. **5**, the first heating part **170** includes a first region **170a** and a second region **170b** that is closer to an outer periphery **179** of the first heating part **170** than is the first region **170a**, as viewed from the Z-axis direction. A planar shape of the first heating part **170** is, for example, a quadrangle such as a rectangle or a square.

(53) The first region **170a** includes a center of the first heating part **170** as viewed from the Z-axis direction. A planar shape of the first region **170a** is, for example, a quadrangle such as a rectangle or a square. In the illustrated example, the planar shape of the first region **170a** is a square. A size

of the first region **170a** in the X-axis direction and a size of the first region **170a** in the Y-axis direction are, for example, about 150 mm.

(54) The first region **170a** does not include the heater **176**. In the example illustrated in FIG. 2, the first region **170a** includes the heat insulating member **172**, the lower plate **174**, and the upper plate **178**. The heater **176** does not constitute the first region **170a**. In the first region **170a**, a gap **2** is formed between the lower plate **174** and the upper plate **178**. A planar shape of the gap **2** is the same as the planar shape of the first region **170a**.

(55) The second region **170b** is adjacent to the first region **170a**. As illustrated in FIG. 5, the second region **170b** surrounds the first region **170a** as viewed from the Z-axis direction. The second region **170b** includes the outer periphery **179**. A size of the second region **170b** in the X-axis direction and a size of the second region **170b** in the Y-axis direction are, for example, about 300 mm. The second region **170b** includes the heater **176**. In the example illustrated in FIG. 2, the second region **170b** includes the heat insulating member **172**, the lower plate **174**, the heater **176**, and the upper plate **178**.

(56) At the time of shaping, a temperature of the second region **170b** is higher than a temperature of the first region **170a**. That is, a temperature of a surface of the second region **170b** on a shaping surface **22** side is higher than a temperature of a surface of the first region **170a** on the shaping surface **22** side. In the illustrated example, the surface of the first region **170a** on the shaping surface **22** side and the surface of the second region **170b** on the shaping surface **22** side are constituted by the upper plate **178**.

(57) 1.1.2.2. Second Heating Part

(58) As illustrated in FIG. 2, the heating part **50** includes the second heating part **180**. The second heating part **180** is provided at an upper side of the shaping surface **22**. The second heating part **180** is supported by the support member **40**. The second heating part **180** moves in conjunction with the nozzle **160**.

(59) The second heating part **180** is provided at an upper side of a position of the nozzle opening **164** at the time of shaping. The second heating part **180** is provided with a through hole **182**. The through hole **182** penetrates the second heating part **180** in the Z-axis direction. At the time of shaping, the nozzle **160** is positioned in the through hole **182**.

(60) The second heating part **180** covers at least a part of the shaping surface **22** when the nozzle opening **164** is positioned at the center of the stage **20** as viewed from the Z-axis direction. When the nozzle opening **164** is positioned at the center of the stage **20** as viewed from the Z-axis direction, the second heating part **180** may cover only a part of the shaping surface **22** or may cover the entire shaping surface **22**. The second heating part **180** heats the shaping surface **22**.

(61) The second heating part **180** includes, for example, a heater **184** and a heat insulating member **186**.

(62) The heater **184** faces the shaping surface **22**. The heater **184** is provided between the shaping surface **22** and the heat insulating member **186**. The heater **184** is, for example, a plate-shaped heater plate. As the heater **184**, for example, a rubber heater is used. The heater **184** heats the shaping surface **22**.

(63) The heat insulating member **186** is provided on the heater **184**. The heat insulating member **186** is coupled to the support member **40**. A shape of the heat insulating member **186** is, for example, a plate shape. The heat insulating member **186** can reduce the heat of the heater **184** transferred to the upper side of the heat insulating member **186**.

(64) Here, FIG. 6 is a perspective view schematically illustrating the second heating part **180**. As illustrated in FIG. 6, the second heating part **180** includes a first region **180a** and a second region **180b** that is closer to an outer periphery **188** of the second heating part **180** than is the first region **180a**, as viewed from the Z-axis direction. A planar shape of the first heating part **180** is, for example, a quadrangle such as a rectangle or a square.

(65) As illustrated in FIGS. 2 and 6, the heater **184** includes, for example, a first portion **184a** and a

second portion **184b**. The first portion **184a** is the heater **184** in the first region **180a**. The second portion **184b** is the heater **184** in the second region **180b**. In the illustrated example, the first portion **184a** and the second portion **184b** are coupled to each other.

(66) In the illustrated example, the first region **180a** includes the first portion **184a** and the heat insulating member **186**. The second region **180b** includes the second portion **184b** and the heat insulating member **186**.

(67) A planar shape of the first region **180a** is, for example, a quadrangle such as a rectangle or a square. In the example illustrated in FIG. 6, the planar shape of the first region **180a** is a square. The planar shape of the first region **180a** is the same as a planar shape of the first portion **184a**. A size of the first region **180a** in the X-axis direction and a size of the first region **180a** in the Y-axis direction are, for example, about 300 mm. The through hole **182** is formed in the first region **180a**.

(68) The second region **180b** is adjacent to the first region **180a**. As illustrated in FIG. 6, the second region **180b** surrounds the first region **180a** as viewed from the Z-axis direction. A planar shape of the second region **180b** is the same as a planar shape of the second portion **184b**. The second region **180b** includes the outer periphery **188**. A size of the second region **180b** in the X-axis direction is, for example, about 600 mm. A size of the second region **180b** in the Y-axis direction is, for example, about 700 mm.

(69) At the time of shaping, a temperature of the second region **180b** is higher than a temperature of the first region **180a**. That is, a temperature of a surface of the second region **180b** on the shaping surface **22** side is higher than a temperature of a surface of the first region **180a** on the shaping surface **22** side. In the example illustrated in FIG. 2, the surface of the first region **180a** on the shaping surface **22** side and the surface of the second region **180b** on the shaping surface **22** side are constituted by the upper plate **184**.

(70) 1.1.3. Temperature Sensor

(71) As illustrated in FIG. 2, the first temperature sensor **60** is provided, for example, in the -X-axis direction of the heater **176**. The first temperature sensor **60** is supported by a support part (not illustrated). The first temperature sensor **60** detects a temperature of the heater **176**.

(72) The second temperature sensor **62** is supported by the heat insulating member **186**. For example, the second temperature sensor **62** overlaps the first portion **184a** of the heater **184** of the second heating part **180** as viewed from the Z-axis direction. The second temperature sensor **62** detects a temperature of the first portion **184a** of the heater **184**.

(73) The third temperature sensor **64** is supported by the heat insulating member **186**. For example, the third temperature sensor **64** overlaps the second portion **184b** of the heater **184** of the second heating part **180** as viewed from the Z-axis direction. The third temperature sensor **64** detects a temperature of the second portion **184b** of the heater **184**. The temperature sensors **60**, **62**, and **64** are, for example, non-contact thermometers.

(74) 1.1.4. Control Part

(75) The control part **70** is implemented by, for example, a computer including a processor, a main storage device, and an input and output interface for exchanging a signal with an outside. The control part **70** implements various functions by, for example, the processor executing a program loaded into the main storage device. Specifically, the control part **70** controls the discharge part **10**, the moving mechanism **30**, and the heating part **50**. The control part **70** may be implemented by a combination of a plurality of circuits instead of the computer.

(76) Here, FIG. 7 is a flowchart illustrating processing of the control part **70**.

(77) First, as illustrated in FIG. 7, as step S1, the control part **70** performs shaping data acquisition processing of acquiring shaping data for shaping a three-dimensional shaped object.

(78) The shaping data includes information on a type of material stored in the material supply part **110**, a movement path of the nozzle **160** with respect to the stage **20**, an amount of shaping material to be discharged from the nozzle **160**, and the like.

(79) The shaping data is created by, for example, causing slicer software installed in a computer

coupled to the three-dimensional shaping device **100** to read shape data. The shape data is data representing a target shape of the three-dimensional shaped object created using three-dimensional computer aided design (CAD) software, three-dimensional computer graphics (CG) software, and the like. As the shape data, for example, data in a standard triangulated language (STL) format or an additive manufacturing file format (AMF) is used. The slicer software divides the target shape of the three-dimensional shaped object into layers having a predetermined thickness, and creates shaping data for each layer. The shaping data is represented by a G code, an M code, or the like. The control part **70** acquires the shaping data from the computer coupled to the three-dimensional shaping device **100** or from a recording medium such as a universal serial bus (USB) memory.

(80) Next, as step **S2**, the control part **70** performs shaping-layer formation processing of forming a shaping layer by discharging the shaping material onto the shaping surface **22** of the stage **20**.

(81) Specifically, the control part **70** plasticizes the material supplied to between the flat screw **130** and the barrel **140** to generate the shaping material, and discharges the shaping material from the nozzle **160**. For example, the control part **70** continuously generates the shaping material until the shaping-layer formation processing is completed.

(82) Further, the control part **70** controls the first heating part **170** to make the temperature of the second region **170b** higher than the temperature of the first region **170a**. For example, the control part **70** controls an output of the heater **176** based on a detection value of the first temperature sensor **60**.

(83) The control part **70** controls the second heating part **180** to make the temperature of the second region **180b** higher than the temperature of the first region **180a**. The control part **70** can individually control an output of the first portion **184a** and an output of the second portion **184b** of the heater **184**. For example, the control part **70** performs control such that the output of the second portion **184b** of the heater **184** is larger than the output of the first portion **184a** of the heater **184**. For example, the control part **70** performs control such that a power density of the second portion **184b** of the heater **184** is higher than a power density of the first portion **184a** of the heater **184**. For example, the control part **70** controls the output of the first portion **184a** based on a detection value of the second temperature sensor **62**. For example, the control part **70** controls the output of the second portion **184b** based on a detection value of the third temperature sensor **64**.

(84) FIG. **8** is a cross-sectional view illustrating the shaping-layer formation processing.

(85) As illustrated in FIG. **8**, the control part **70** controls the moving mechanism **30** to change the relative position between the nozzle **160** and the stage **20** based on the acquired shaping data, and controls the discharge part **10** to discharge the shaping material from the nozzle **160** toward the stage **20**.

(86) Specifically, before the shaping-layer formation processing is started, that is, before formation of a shaping layer **L1** that is a first shaping layer is started, the nozzle **160** is disposed at an initial position in the $-X$ axis direction with respect to an end portion of the stage **20** in the $-X$ axis direction. When the shaping-layer formation processing is started, as illustrated in FIG. **8**, the control part **70** controls the moving mechanism **30** to move the nozzle **160** in the $+X$ -axis direction relative to the stage **20**, for example. When the nozzle **160** passes over the stage **20**, the shaping material is discharged from the nozzle **160**. Accordingly, the shaping layer **L1** is formed. In FIG. **8**, n is any natural number, and layers up to the n -th shaping layer **Ln** are illustrated.

(87) Next, as illustrated in FIG. **7**, as step **S3**, the control part **70** performs determination processing of determining, based on the shaping data, whether the formation of all the shaping layers is completed.

(88) When it is determined that the formation of all the shaping layers is not completed (“NO” in step **S3**), the control part **70** returns the processing to step **S2**. The control part **70** repeats step **S2** and step **S3** until it is determined in step **S3** that the formation of all the shaping layers is completed.

(89) On the other hand, when it is determined that the formation of all the shaping layers is

completed (“YES” in step S3), the control part **70** ends the processing.

(90) 1.1.5. Operation and Effect

(91) In the three-dimensional shaping device **100**, the heating part **50** includes the first region **180a** and the second region **180b** closer to the outer periphery **188** of the heating part **50** than is the first region **180a** as viewed from the Z-axis direction, and the control part **70** controls the heating part **50** to make the temperature of the second region **180b** higher than the temperature of the first region **180a**.

(92) Therefore, in the three-dimensional shaping device **100**, for example, as compared with a case where the temperature of the second region is the same as the temperature of the first region, the temperature of the shaping material deposited in the second region **180b** becoming lower than the temperature of the shaping material deposited in the first region **180a** due to the shaping material deposited in the second region **180b** being cooled by the outside air can be prevented. Accordingly, it is possible to lower the possibility of occurrence of warpage or deformation in the shaping material deposited in the second region **180b**.

(93) In the three-dimensional shaping device **100**, the heating part **50** includes the first heating part **170** provided at the lower side of the nozzle opening **164**, the first heating part **170** heats the stage **20**, the second region **170b** of the first heating part **170** includes the heater **176**, and the first region **170a** of the first heating part **170** does not include the heater **176**. Therefore, in the three-dimensional shaping device **100**, the temperature of the second region **170b** of the first heating part **170** can be made higher than the temperature of the first region **170a** of the first heating part **170**.

(94) In the three-dimensional shaping device **100**, the heating part **50** includes the second heating part **180** provided at the upper side of the position of the nozzle opening **164** at the time of shaping. The second heating part **180** moves in conjunction with the nozzle opening **164**, and covers at least a part of the shaping surface **22** when the nozzle opening **164** is positioned at the center of the stage **20** as viewed from the Z-axis direction. Therefore, in the three-dimensional shaping device **100**, at least a part of the shaping surface **22** can be heated by the second heating part **180**.

(95) In the three-dimensional shaping device **100**, the first region **180a** and the second region **180b** of the second heating part **180** include the heater **184**, and the control part **70** performs control so that the output of the heater **184** in the second region **180b** is larger than the output of the heater **184** in the first region **180a**. Therefore, in the three-dimensional shaping device **100**, the temperature of the second region **180b** of the second heating part **180** can be made higher than the temperature of the first region **180a** of the second heating part **180**.

(96) The three-dimensional shaping device **100** includes the first temperature sensor **60** that detects the temperature of the heater **176**, and the control part **70** controls the output of the heater **176** based on the detection value of the first temperature sensor **60**. Therefore, in the three-dimensional shaping device **100**, the output of the heater **176** can be controlled based on the temperature of the heater **176**.

(97) In the three-dimensional shaping device **100**, the first heating part **170** includes the lower plate **174** and the upper plate **178**, and the heater **176** as a heater plate is fixed by being sandwiched between the lower plate **174** and the upper plate **178**. Therefore, in the three-dimensional shaping device **100**, since the heater **176** is not fixed by a bonding agent, the warpage of the heater **176** due to the bimetal can be reduced.

2. Second Embodiment

(98) 2.1. Three-Dimensional Shaping Device

(99) Next, a three-dimensional shaping device according to a second embodiment will be described with reference to the drawings. FIG. **9** is a perspective view schematically illustrating a three-dimensional shaping device **200** according to the second embodiment. Hereinafter, in the three-dimensional shaping device **200** according to the second embodiment, members having the same functions as those of the constituent members of the three-dimensional shaping device **100** described above are denoted by the same reference signs, and a detailed description thereof will be

omitted.

(100) As illustrated in FIG. 9, the three-dimensional shaping device **200** is different from the three-dimensional shaping device **100** described above in that a thickness of a fourth portion **186b** of the heat insulating member **186** is larger than a thickness of a third portion **186a**. In the illustrated example, the “thickness” refers to a size in the Z-axis direction.

(101) The heat insulating member **186** includes the third portion **186a** and the fourth portion **186b**. A planar shape of the third portion **186a** is, for example, the same as the planar shape of the first portion **184a**. The planar shape of the fourth portion **186b** is, for example, the same as the planar shape of the second portion **184b**. In the illustrated example, the first region **180a** includes the first portion **184a** and the third portion **186a**. The second region **180b** includes the second portion **184b** and the fourth portion **186b**. The third portion **186a** is the heat insulating member **186** in the first region **180a**. The fourth portion **186b** is the heat insulating member **186** in the second region **180b**. In the illustrated example, the third portion **186a** and the fourth portion **186b** are coupled to each other.

(102) In the three-dimensional shaping device **200**, a thickness of the heat insulating member **186** in the second region **180b** is larger than a thickness of the heat insulating member **186** in the first region **180a**. Therefore, in the three-dimensional shaping device **200**, for example, as compared with a case where the thickness of the heat insulating member in the second region is the same as the thickness of the heat insulating member in the first region, a temperature of the second region **180b** of the second heating part **180** can be made higher than a temperature of the first region **180a** of the second heating part **180**. For example, even if the output of the heater **184** in the second region **180b** is not made larger than the output of the heater **184** in the first region **180a** by the control part **70**, the temperature of the second region **180b** can be made higher than the temperature of the first region **180a**.

(103) Although not illustrated, a thickness of the heat insulating member **172** in the second region **170b** of the first heating part **170** may be larger than a thickness of the heat insulating member **172** in the first region **170a** of the first heating part **170**.

(104) 2.2. Modification

(105) Next, a three-dimensional shaping device according to a modification of the second embodiment will be described with reference to the drawings. FIG. 10 is a cross-sectional view schematically illustrating a three-dimensional shaping device **210** according to the modification of the second embodiment. Hereinafter, in the three-dimensional shaping device **210** according to the modification of the second embodiment, members having the same functions as those of the constituent members of the three-dimensional shaping devices **100** and **200** described above are denoted by the same reference signs, and a detailed description thereof will be omitted.

(106) As illustrated in FIG. 10, the three-dimensional shaping device **210** is different from the three-dimensional shaping device **200** described above in that thermal conductivity of the fourth portion **186b** is lower than thermal conductivity of the third portion **186a**.

(107) As the third portion **186a**, for example, an INSULTEX™ (registered trademark) cloth is used. As the fourth portion **186b**, for example, a ROSLIM™ board is used. A thickness of the third portion **186a** and a thickness of the fourth portion **186b** are, for example, the same. Although not illustrated, the thickness of the fourth portion **186b** may be larger or smaller than the thickness of the third portion **186a**.

(108) In the three-dimensional shaping device **210**, the heating part **50** includes the heat insulating member **186** and the heater **184** as a heater plate provided between the heat insulating member **186** and the shaping surface **22**, and thermal conductivity of the heat insulating member **186** in the second region **180b** is lower than thermal conductivity of the heat insulating member **186** in the first region **180a**. Therefore, in the three-dimensional shaping device **210**, a temperature of the second region **180b** of the second heating part **180** can be made higher than a temperature of the first region **180a** of the second heating part **180**, as compared with a case where the thermal

conductivity of the heat insulating member in the second region is the same as the thermal conductivity of the heat insulating member in the first region. For example, even if an output of the heater **184** in the second region **180b** is not made larger than an output of the heater **184** in the first region **180a** by the control part **70**, the temperature of the second region **180b** can be made higher than the temperature of the first region **180a**.

(109) Thermal conductivity of the heat insulating member **172** in the second region **170b** of the first heating part **170** may be lower than thermal conductivity of the heat insulating member **172** in the first region **170a** of the first heating part **170**.

3. Third Embodiment

(110) 3.1. Three-Dimensional Shaping Device

(111) Next, a three-dimensional shaping device according to a third embodiment will be described with reference to the drawings. FIG. **11** is a cross-sectional view schematically illustrating a three-dimensional shaping device **300** according to the third embodiment. Hereinafter, in the three-dimensional shaping device **300** according to the third embodiment, members having the same functions as those of the constituent members of the three-dimensional shaping device **100** described above are denoted by the same reference signs, and a detailed description thereof will be omitted.

(112) In the three-dimensional shaping device **300**, as illustrated in FIG. **11**, a configuration of the second heating part **180** is different from that of the three-dimensional shaping device **100** described above.

(113) As illustrated in FIG. **11**, the second heating part **180** of the three-dimensional shaping device **300** includes, for example, a lower plate **380**, the heater **184**, an upper plate **382**, the heat insulating member **186**, a reinforcing member **384**, and a radiant heat suppressing member **386**.

(114) The lower plate **380** faces the shaping surface **22** of the stage **20**. A material of the lower plate **380** is, for example, aluminum. For example, an oxide film is provided on an upper surface and a lower surface of the lower plate **380**. With the oxide film, radiant heat from the heater **184** can be easily accumulated, and the shaping material deposited on the shaping surface **22** can be efficiently heated.

(115) The heater **184** is provided on the lower plate **380**. The heater **184** is provided between the lower plate **380** and the upper plate **382**. The heater **184** is fixed by being sandwiched between the lower plate **380** and the upper plate **382**. The heater **184** is not bonded to the lower plate **380** with a bonding agent. The heater **184** is not bonded to the upper plate **382** with a bonding agent. The heater **184** is fixed by its own weight or a load of the upper plate **382**.

(116) The upper plate **382** is provided on the heater **184**. The upper plate **382** is provided between the heater **184** and the heat insulating member **186**. A material of the upper plate **382** is, for example, aluminum. An upper surface and a lower surface of the upper plate **382** are, for example, polished mirror surfaces. Accordingly, the radiant heat from the heater **184** can be reflected to the shaping surface **22** side.

(117) The heat insulating member **186** is provided on the upper plate **382**. The heat insulating member **186** is provided between the upper plate **382** and the radiant heat suppressing member **386**. For example, the heat insulating member **186** is cotton-shaped. By using the cotton-shaped heat insulating member **186**, weight reduction can be achieved as compared with a case of using, for example, a plate-shaped heat insulating member. As the heat insulating member **186**, for example, alkaline earth silicate (AES) wool is used.

(118) The reinforcing member **384** is provided on the upper plate **382**. The reinforcing member **384** is provided between the upper plate **382** and the radiant heat suppressing member **386**. The reinforcing member **384** is coupled to the upper plate **382** and the radiant heat suppressing member **386**. A material of the reinforcing member **384** is, for example, steel use stainless (SUS). The reinforcing member **384** reinforces the second heating part **180**.

(119) The radiant heat suppressing member **386** is provided on the heat insulating member **186** and

the reinforcing member **384**. As the radiant heat suppressing member **386**, for example, an aluminum foil is used. The heater **184** is provided between the shaping surface **22** and the radiant heat suppressing member **386**. The radiant heat suppressing member **386** suppresses the radiant heat from the heater **184** from being transferred to an opposite side with respect to the shaping surface **22**. A lower surface of the radiant heat suppressing member **386** is, for example, a polished mirror surface. Accordingly, the radiant heat suppressing member **386** can reflect the radiant heat from the heater **184** to the shaping surface **22** side. An oxide film may be provided on an upper surface of the radiant heat suppressing member **386**.

(120) In the three-dimensional shaping device **300**, the heating part **50** includes the lower plate **380** and the upper plate **382**, and the heater **184** as a heater plate is fixed by being sandwiched between the lower plate **380** and the upper plate **382**. Therefore, in the three-dimensional shaping device **300**, since the heater **184** is not fixed with a bonding agent, the warpage of the heater **184** due to bimetal can be reduced.

(121) In the three-dimensional shaping device **300**, the heating part **50** includes the radiant heat suppressing member **386** that suppresses the radiant heat from the heater **184** serving as a heater plate from being transferred to the opposite side with respect to the shaping surface **22**, and the heater **184** is provided between the shaping surface **22** and the radiant heat suppressing member **386**. Therefore, with the three-dimensional shaping device **300**, it is possible to efficiently heat the shaping material deposited on the shaping surface **22** as compared with a case where the radiation heat suppressing member is not provided.

(122) Although not illustrated, the first heating part **170** may include a radiant heat suppressing member that suppresses radiant heat from the heater **176** from being transferred to the opposite side with respect to the shaping surface **22**.

(123) 3.2. Modification

(124) Next, a three-dimensional shaping device according to a modification of the third embodiment will be described with reference to the drawings. FIG. **12** is a cross-sectional view schematically illustrating a three-dimensional shaping device **310** according to the modification of the third embodiment. Hereinafter, in the three-dimensional shaping device **310** according to the modification of the third embodiment, members having the same functions as those of the constituent members of the three-dimensional shaping devices **100** and **300** described above are denoted by the same reference signs, and a detailed description thereof will be omitted.

(125) In the three-dimensional shaping device **310**, as illustrated in FIG. **12**, a configuration of the second heating part **180** is different from that of the three-dimensional shaping device **300** described above.

(126) As illustrated in FIG. **12**, the second heating part **180** of the three-dimensional shaping device **310** includes, for example, the lower plate **380**, the heater **184**, the radiant heat suppressing member **386**, the heat insulating member **186**, and a cover **388**.

(127) The heater **184** is provided between the lower plate **380** and the radiant heat suppressing member **386**. The radiant heat suppressing member **386** is provided on the heater **184**. The radiant heat suppressing member **386** is provided between the heater **184** and the heat insulating member **186**. The heat insulating member **186** is provided on the radiant heat suppressing member **386**. The heat insulating member **186** is provided between the radiant heat suppressing member **386** and the cover **388**. A shape of the heat insulating member **186** is a plate shape. As the heat insulating member **186**, for example, a ROSLIM™ board is used. The cover **388** is provided on the heat insulating member **186**. A material of the cover **388** is, for example, plastic.

4. Fourth Embodiment

(128) Next, a three-dimensional shaping device according to a fourth embodiment will be described.

(129) Hereinafter, with respect to the three-dimensional shaping device according to the fourth embodiment, points different from the example of the three-dimensional shaping device **100**

according to the first embodiment described above will be described, and a description of the same points will be omitted.

(130) In the three-dimensional shaping device **100** described above, the material supplied from the material supply part **110** is the ABS resin.

(131) Meanwhile, in the three-dimensional shaping device according to the fourth embodiment, the material supplied from the material supply part **110** is a material other than the ABS resin or a material obtained by adding another component to the ABS resin.

(132) Examples of the material supplied from the material supply part **110** include materials having various materials such as a thermoplastic material, a metal material, and a ceramic material as a main material. Here, the “main material” means a material serving as a main component forming a shape of a shaped object, and means a material having a content of 50% by mass or more in the shaped object. The materials described above include those acquired by melting these main materials alone, and those acquired by melting a part of components contained together with the main materials into a paste form.

(133) As the thermoplastic material, for example, a thermoplastic resin can be used. Examples of the thermoplastic resin include general-purpose engineering plastics and super engineering plastics.

(134) Examples of the general-purpose engineering plastic include polypropylene (PP), polyethylene (PE), polyacetal (POM), polyvinyl chloride (PVC), polyamide (PA), polylactic acid (PLA), polyphenylene sulfide (PPS), polycarbonate (PC), modified polyphenylene ether, polybutylene terephthalate, and polyethylene terephthalate.

(135) Examples of the super engineering plastic include polysulfone (PSU), polyethersulfone (PES), polyphenylene sulfide (PPS), polyarylate (PAR), polyimide (PI), polyamideimide (PAI), polyetherimide (PEI), and polyether ether ketone (PEEK).

(136) Additives such as a pigment, a metal, a ceramic, a wax, a flame retardant, an antioxidant, and a heat stabilizer may be mixed into the thermoplastic material. In the plasticizing part **120**, the thermoplastic material is plasticized and converted into a molten state by the rotation of the flat screw **130** and the heating of the barrel heater **150**. The shaping material generated in this way is deposited from the nozzle **160** and then cured due to a decrease in temperature. The thermoplastic material may be discharged from the nozzle **160** in a state in which the material is heated to a temperature equal to or higher than a glass transition point thereof and is melted completely.

(137) In the plasticizing part **120**, for example, the metal material may be used as the main material instead of the thermoplastic material described above. In this case, a component to be melted at the time of generating the shaping material may be mixed into a powder material obtained by converting the metal material into a powder, and then the mixture is fed into the plasticizing part **120**.

(138) Examples of the metal material include a single metal such as magnesium (Mg), iron (Fe), cobalt (Co), chromium (Cr), aluminum (Al), titanium (Ti), copper (Cu), and nickel (Ni), or an alloy containing one or more of these metals, or maraging steel, stainless steel, a cobalt-chromium-molybdenum alloy, a titanium alloy, a nickel alloy, an aluminum alloy, a cobalt alloy, and a cobalt-chromium alloy.

(139) In the plasticizing part **120**, the ceramic material may be used as the main material instead of the metal material described above. Examples of the ceramic material include an oxide ceramic such as silicon dioxide, titanium dioxide, aluminum oxide, and zirconium oxide, and a non-oxide ceramic such as aluminum nitride.

(140) A powder material of the metal material or the ceramic material supplied from the material supply part **110** may be a mixed material in which a plurality of types of powder of the single metals or powder of the alloys are mixed, or a mixed material in which a plurality of types of powder of the ceramic materials are mixed. In addition, the powder material of the metal material or the ceramic material may be coated with, for example, the above-described thermoplastic resin or another thermoplastic resin. In this case, the thermoplastic resin may be melted to exhibit

fluidity in the plasticizing part **120**.

(141) For example, a solvent may be added to the powder material of the metal material or the ceramic material supplied from the material supply part **110**. Examples of the solvent include: water; (poly)alkylene glycol monoalkyl ethers such as ethylene glycol monomethyl ether, ethylene glycol monoethyl ether, propylene glycol monomethyl ether, and propylene glycol monoethyl ether; acetate esters such as ethyl acetate, n-propyl acetate, isopropyl acetate, n-butyl acetate, and iso-butyl acetate; aromatic hydrocarbons such as benzene, toluene, and xylene; ketones such as methyl ethyl ketone, acetone, methyl isobutyl ketone, ethyl-n-butyl ketone, diisopropyl ketone, and acetylacetone; alcohols such as ethanol, propanol, and butanol; tetraalkylammonium acetates; sulfoxide-based solvents such as dimethyl sulfoxide and diethyl sulfoxide; pyridine-based solvents such as pyridine, γ -picoline, and 2,6-lutidine; tetraalkylammonium acetates (such as tetrabutylammonium acetate); and ionic liquids such as butyl carbitol acetate.

(142) In addition, for example, a binder may be added to the powder material of the metal material or the ceramic material supplied from the material supply part **110**. Examples of the binder include an acrylic resin, an epoxy resin, a silicone resin, a cellulose-based resin, other synthetic resins, PLA, PA, PPS, PEEK, and other thermoplastic resins.

(143) The embodiments and the modifications described above are merely examples, and the present disclosure is not limited thereto. For example, the embodiments and the modifications can be appropriately combined.

(144) The present disclosure includes a configuration substantially the same as the configurations described in the embodiments, for example, a configuration having the same functions, methods, and results, or a configuration having the same objects and effects. In addition, the present disclosure includes a configuration in which a non-essential portion of the configurations described in the embodiments is replaced. In addition, the present disclosure includes a configuration having the same operation and effect as the configurations described in the embodiments, or a configuration capable of achieving the same object. Further, the present disclosure includes a configuration obtained by adding a known technique to the configurations described in the embodiments.

(145) The following contents are derived from the above-described embodiments and modifications.

(146) One aspect of a three-dimensional shaping device includes: a discharge part that includes a nozzle configured to discharge a shaping material from a nozzle opening thereof; a stage that includes a shaping surface on which the shaping material is to be deposited; a heating part configured to heat the shaping material deposited on the stage; and a control part configured to control the heating part, the heating part includes a first region and a second region closer to an outer periphery of the heating part than is the first region, as viewed from a direction of a perpendicular line of the shaping surface, and the control part controls the heating part to make a temperature of the second region higher than a temperature of the first region.

(147) According to the three-dimensional shaping device, it is possible to lower the possibility of occurrence of warpage or deformation in the shaping material deposited in the second region.

(148) In one aspect of the three-dimensional shaping device, the heating part may include a first heating part provided at a lower side of the nozzle opening, the first heating part may heat the stage, the second region of the first heating part may be provided with a heater, and the first region of the first heating part may not be provided with a heater.

(149) According to the three-dimensional shaping device, a temperature of the second region of the first heating part can be made higher than a temperature of the first region of the first heating part.

(150) In one aspect of the three-dimensional shaping device, the heating part may include a second heating part provided at an upper side of a position of the nozzle opening at the time of shaping, and the second heating part may move in conjunction with the nozzle opening, and cover at least a part of the shaping surface when the nozzle opening is positioned at a center of the stage as viewed

from the direction of the perpendicular line.

(151) According to the three-dimensional shaping device, at least a part of the shaping surface can be heated by the second heating part.

(152) In one aspect of the three-dimensional shaping device, the first region and the second region of the second heating part may be both provided with a heater, and the control part may perform control such that an output of the heater in the second region is larger than an output of the heater in the first region.

(153) According to the three-dimensional shaping device, a temperature of the second region of the second heating part can be made higher than a temperature of the first region of the second heating part.

(154) The one aspect of the three-dimensional shaping device may further include a temperature sensor configured to detect a temperature of the heater, and the control part may control an output of the heater based on a detection value of the temperature sensor.

(155) According to the three-dimensional shaping device, the output of the heater can be controlled based on the temperature of the heater.

(156) In one aspect of the three-dimensional shaping device, the heating part may include a heat insulating member, and a heater plate provided between the heat insulating member and the shaping surface, and a thickness of the heat insulating member in the second region may be larger than a thickness of the heat insulating member in the first region.

(157) According to the three-dimensional shaping device, the temperature of the second region of the second heating part can be made higher than the temperature of the first region of the second heating part.

(158) In one aspect of the three-dimensional shaping device, the heating part may include a heat insulating member, and a heater plate provided between the heat insulating member and the shaping surface, and thermal conductivity of the heat insulating member in the second region may be lower than thermal conductivity of the heat insulating member in the first region.

(159) According to the three-dimensional shaping device, the temperature of the second region of the second heating part can be made higher than the temperature of the first region of the second heating part.

(160) In one aspect of the three-dimensional shaping device, the heating part may include a first plate and a second plate, and the heater plate may be fixed by being sandwiched between the first plate and the second plate.

(161) According to the three-dimensional shaping device, warpage of the heater due to bimetal can be reduced.

(162) In one aspect of the three-dimensional shaping device, the heating part may include a radiant heat suppressing member configured to suppress radiant heat from the heater plate from being transferred to an opposite side with respect to the shaping surface, and the heater plate may be provided between the shaping surface and the radiant heat suppressing member.

(163) According to the three-dimensional shaping device, it is possible to efficiently heat the shaping material deposited on the shaping surface.

Claims

1. A three-dimensional shaping device that is a pellet three-dimensional extrusion printer, the three-dimensional shaping device comprising: a discharge part that includes a nozzle configured to discharge a shaping material from a nozzle opening thereof; a stage that includes a shaping surface on which the shaping material is to be deposited; a first heating part disposed at a lower side of the nozzle opening and configured to heat the stage to heat the shaping material deposited on the stage, the first heating part including a first plate, a second plate, and a first heater that is a plate-shaped rubber heater, the first heater being disposed between the first plate and the second plate such that a

gap is formed by the first heater, the first plate, and the second plate in a first region of the first heating part, and such that the first heater is disposed in a second region of the first heating part, the second region being closer to an outer periphery of the first heating part than the first region, the outer periphery of the first heating part being an outermost edge of the first heating part as viewed from a perpendicular direction perpendicular to the shaping surface; and a control part including a processor and a main storage device, electrically connected to the discharge part and the first heating part, and executing a program loaded into the main storage device to control the discharge part and the first heating part to perform a shaping-layer formation processing that includes forming a shaping layer by discharging the shaping material onto the stage, and includes making a temperature of the second region higher than a temperature of the first region.

2. The three-dimensional shaping device according to claim 1, further comprising a second heating part disposed at an upper side of a position of the nozzle opening at the time of shaping, wherein the control unit further controls the second heating part to move the second heating part in conjunction with the nozzle opening so as to cover at least a part of the shaping surface when the nozzle opening is positioned at a center of the stage as viewed from the perpendicular direction.

3. The three-dimensional shaping device according to claim 2, wherein the second heating part including a second heater that is a plate-shaped rubber heater and is disposed in both a third region of the second heating part and a fourth region of the second heating part, the fourth region being closer to an outer periphery of the second heating part than the third region, the outer periphery of the second heating part being an outermost edge of the second heating part as viewed from the perpendicular direction, and the control part controls the second heating part such that an output of the second heater in the fourth region is larger than an output of the second heater in the third region.

4. The three-dimensional shaping device according to claim 1, further comprising: a temperature sensor configured to detect a temperature of the first heater, wherein the control part controls the first heating portion to control an output of the first heater based on a detection value of the temperature sensor.

5. The three-dimensional shaping device according to claim 1, wherein the first heater is fixed to the first plate and the second plate.

6. The three-dimensional shaping device according to claim 1, further comprising a second heating part that is disposed at an upper side of a position of the nozzle opening at the time of shaping, wherein the second heating part includes a second heater that is a plate-shaped rubber heater, and a radiant heat suppressing member that is configured to suppress radiant heat from the second heater from being transferred to an opposite side with respect to the shaping surface, and the second heater is disposed between the shaping surface and the radiant heat suppressing member.
